Corncob Redux - Strategy and Flying Hints

Lost Engines - Corncob Redux
Reviewed edition of the original game documentation.
Controls and changed rules describe Redux defaults.
Original credits, source tables and archival details are retained.

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General strategy and flying hints

Attack Priorities:

When attacking an installation protected by both KLA's and AAA, first attack the high-quality AAA. These defenses will chew you up the most quickly while undergoing evasive maneuvers. Then go for the top-notch KLA sites. Once these threats have been dealt with, proceed with the rest of the defenses at your leisure.

Enemy activity at a distance (Redux):

Enemies, projectiles and production sites continue to update when you leave their area. Travelling away no longer freezes their timers or repairs their damage. Weapon ranges and local production caps still apply; draw distance does not change these rules.

Use the half barrel roll to ease blackouts quickly:

When a high-g turn blacks you out, the quickest way to get that blood back into your brain is to do a quick high-negative-g turn. Simply easy up on the stick, and flip your plane around upside down, and push forward
on the stick. If you managed to flip the plane correctly, you should still be turning in the desired direction, and your vision should return very quickly.

Use dumpy planes to do high-risk platform missions:

When you have to land on a suspended airstrip and proceed on foot to a difficult assignment, it is not unusual to fall off a platform, and to leave your plane abandoned up on the airstrip. You may want to ferry your late model planes away from one of the airstrips in the theater.
Then when you fly again from the planeless airstrip, the program will ask
you if you want to pull a dumpy plane from storage. Take the dumpy plane
to the suspended mission, so you won't lose a good plane if you fall.

Avoid bounce-back on suicide missions:

Some missions involve suiciding your plane into a generator orb. Be aware that your plane will bounce backward upon impact. It is not uncommon
for the plane to bounce back toward you so that you find yourself back in the cockpit after collision, hurtling toward the ground in a ball of flaming wreckage.
Be ready to eject again in case this occurs. Take care to time your ejection correctly so this doesn't happen.
Note also, the hint given above about using dumpy planes for dangerous mission applies to suicide missions also.

Flying low over craters (Redux):

Bomb explosions and their debris remain dangerous. The lingering crater cannot hurt you or your aircraft, so it no longer creates an invisible obstacle above the ground. Its blast damage against enemy structures is retained.

General strategy and flying hints (continued)

Use rudder to reduce airspeed on runway:

  Sometimes when landing on suspended runways it is imperative to come to
a quick stop. Hold down the brake key, and simultaneously work the rudder
back and forth. The swishing of the tail will help you slow down, as
well as making your path longer to the end of the runway. Take care not
to direct the plane off the edge.

Use barrel roll to reduce airspeed on approach:

  When making a high-speed landing approach, you sometimes want to drop
your excess speed. Dropping the flaps, and cutting back on the throttle
many times isn't a big enough effect, especially if your angle of descent
is too steep. In this case, try pulling the stick all the way back, and
fully to the side. The plane will travel in a barrel roll and end up in
approximately in the same flight path, but with the airspeed down
significantly.

Techniques for escaping with your plane intact

Use teleportation to get out of trouble:

  This is a bit of a cheat, since it is an unrealistic way out of impending
doom. The purists will want to not do this, since it twists a convenience
feature of the game into an irrational combat advantage.
  If you are in deep doo-doo and are worried that you will not get out
of this situation alive, simply hit F2, and select a new mission. If you
are somewhat damaged, you may want to choose a mission which is close
to your airstrip. Also, you might want to choose a mission which you have
already completed.
  Note that to keep this feature from being abused, there is a rule that
you cannot teleport if your plane has sustained serious damage.

Use autopilot when heavily damaged:

  When your plane is a flying scrap, and you are having difficulty
keeping your plane in the air due to excessive control damage, try turning
on the autopilot. This should help you keep the plane reasonably level.
The autopilot will react faster than you, and can make the difference needed
to make it home safely.
  Beware that the autopilot will interfere with attempts at evasive maneuvers.
If you are going to try to dodge a KLA, make sure you disengage the autopilot
first.

Use the emergency rocket booster to full advantage:

General strategy and flying hints (continued)

When you are surrounded by KLA's, and you have taken too many hits to risk continuing, hit 'i' to ignite the rocket booster. If you are at low altitude, pull the nose up to vertical to get the maximum altitude gain from the rocket. However, be sure to push the nose back to the horizontal BEFORE the booster runs out. This is because you will really need the airspeed in addition to the altitude.

If you already have good altitude (for instance if you are in a high altitude mission), then use the rocket booster for pure speed. Disengage the flaps to maximize airspeed, put yourself in level flight, and let her rip. You can outrun almost any of the KLA's with the booster.

Note that it is generally a good idea to get the plane headed towards your airstrip before engaging the rocket. Tight turns with the booster running will hurt your gained airspeed by quite a large amount.

Hints for defeating KLA and AAA

Blast distant airborne KLA's:

When approaching a mission from afar, many times there will be KLA's already airborne and approaching from the front. Frequently, these pre-launched KLA's will be of higher quality than any of the nearby KLA sites. Therefore, it is usually a good idea to blow these approachers out of the sky with missiles and guns. Since they are advancing upon you head-on, it is an easy zero-deflection shot.

When NOT to shoot KLA's:

Keep in mind that KLA sites will only produce KLA's when the number of active KLA's presently flying is less than the number of sites. Therefore if you shoot them, the sites will make more. The key here is to only shoot the high-quality deathballs. There is a good chance that the next site to create a replacement will be a lower quality site. Thus by shooting the high quality KLA, you have traded it for a low quality model.

Of course, for the same reason, you should NEVER shoot a poor quality KLA if there is a variety of sites around. A high quality site may make its replacement.

Remember that the lower quality models pose almost no threat to the seasoned pilot.

Shoot at KLA sites at every chance:

When a KLA site takes some damage, it is prevented from production for a period of time. Therefore, even if you haven't enough time to get in enough hits to destroy the site, even one hit with a bullet is enough to keep it out of commission for tens of seconds.

When you see a clump of KLA sites, always spray a hail of bullets and missiles over them just to stunt their output for a while.

KLA production and time (Redux):

Production follows fixed simulation time, rather than displayed frames.

General strategy and flying hints (continued)

Low frame rates or different rendering resolutions do not slow a site's countdown. The T-key time-advance exploit has been removed. Damaging sites
still delays production, and the local population cap still applies.

Get KLA's to fly into enemy structures

  Corncob was designed so that enemy objects can collide with each other.
A useful technique for dealing with a long string of tailing KLA's is to simply trick them into flying into their own structures.
  For example, it is possible even to blow up a generator orb by leading deathballs into it.
  For best results you will need to keep your airspeed up, and make your 180 degree turns be gentle to avoid dispersing the stream.

Destroying strings of KLA's with your runway:

  If you have used all your bombs, and there is a long string of KLA's about to hit you on your landing approach, there is no need to worry. The materials used in the runway will cause the KLA's to detonate prematurely if they get close.
  Even if you have no intention of landing, a low flyby over the runway will cause the trail to explode as they follow your lead over the runway.

Get rid of a KLA stream by dropping a low bomb:

  Certainly you can take out a single KLA which is very close to your tail
simply by dropping a bomb. You aren't required to do any maneuver, just drop the bomb. However, you will frequently find that you have a long string of them behind you, and only three bombs. By flying low, and dropping a bomb, the debris flying up from the explosion can take out almost the whole string as they attempt to fly through the explosion.

Keep your airspeed up around AAA:

  The higher quality AAA will attempt to lead your plane so that the slow moving shells will arrive on target despite the forward motion of your plane. Thus, the higher your airspeed, the more dispersed the enemy shells will be.
  Letting your airspeed drop due to maneuvers is the fastest way to get blown out of the sky by AAA. When you drop your flaps to mix it up with the KLA's is when they will get a bead on you. Eliminate all the top notch AAA before getting into knife-fights with the deathballs.

Stay low to avoid AAA fire:

  Ground-based AAA batteries cannot fire very low to the ground, for fear of taking out their own structures. It is possible to stay under their firing angle by flying extremely close to the ground. The closer you are to the AAA site, the lower you must be to avoid fire.

Stay level with suspended AAA:

  AAA shells are programmed at firing time to explode at a certain altitude. Since the enemy is attempting to kill you, they program each

General strategy and flying hints (continued)

shell to explode when it reaches the altitude of your plane. The exact altitude of detonation is randomly distributed around the target altitude.

The result is that if you manage to fly level with the AAA, the shells are already at their detonation altitude before they even leave the AAA muzzle! Therefore the shells will almost always detonate before hitting your plane.

Always use bullets and missiles when attacking AAA:

The bad news is that AAA shells will explode your missiles as you fire on an AAA site. Since the AAA site will be sending a continuous stream of shells at you, then it is unlikely for your missiles to make it to the target.

The good news is that bullets will explode approaching AAA shells. Therefore the best technique is to let loose a volley of bullets to take out the slow-moving approaching AAA shells. Then send a group of missiles together with more bullets. That way the faster-moving bullets will take out the AAA shells, clearing the way for the powerful missiles.

Never fly with your tail to the AAA:

AAA works best on zero-deflection targets. If you cannot use your weapons to clear the air of approaching shells, then your best bet is to be flying fast and perpendicular to the AAA. You are most vulnerable when flying directly away from the AAA site. Then the AAA can shoot directly at you with no target-leading or anticipation, and the rate of closure is the slowest to ensure an accurate blast.

Avoid bombing AAA unless desperate:

Many a pilot has been toasted by his own bomb whilst attempting this. The AAA shells will detonate your bomb in the air. For lower quality AAA, the pilot can sometimes get lucky when the firing rate of the site is fairly low.

Tips for downing flying saucers

Keep up the airspeed:

When dealing with saucers with AAA, you must be aware of the fact that you will not be able to down the saucer with one pass. Therefore after passing the saucer, it will chew you up as you attempt to fly away in order to make another pass. Your only defense is fly as fast as possible, so that the time in which your tail is turned toward the saucer is minimized.

In addition, the faster your airspeed, the faster your missiles will reach the target. The launch speed of your missiles affects the acceleration to max velocity in a positive way. With a moving target like a saucer, the faster your ordnance travels the better.

Keep level with the saucer:

General strategy and flying hints (continued)

For the same reason that you should stay level with suspended AAA, you should stay level with AAA fitted saucers. This is especially true when you are pulling away from the saucer.

Try to time the oscillations of the saucers movement:

You will notice that the saucer's evasive maneuvers are very periodic. Therefore if you notice that your missiles are always missing because of the saucer's motion, try shooting where you think the saucer will be when the missiles arrive. If that isn't working, just spread your missiles around the whole area through which the saucer is swinging.

Try to get 'behind' the saucer:

At any moment, all the saucers are traveling in a horizontal direction. You want to approach in the same direction as the saucer is moving. This gives you a slower rate of closure, so that you can get more shots in before you have to peel off to avoid collision. It is possible to down a saucer in one pass using this technique.

Do not fight saucers in close:

Fighting a saucer up close usually ends in disaster. The evasive maneuvers which it is constantly executing make it too unpredictable a target at close range. In addition, there is a great risk of collision. Upon collision with a saucer, your plane becomes a ball of twisted wreckage.

Although collision will take out the saucer, the brass doesn't look at it as an even trade. Perhaps for the desperate, it may be worth suiciding a dumpy plane for a top-ace saucer, though.

Hints for vanquishing ground vehicles:

Keep your distance:

Ground vehicles do much random zipping around. If you are up close, those movements will cause you to go through wild aerodynamic gyrations in order to keep those critters in your sights. This is very likely to end in a fatal stall, since when attacking ground targets you are probably flying too low to recover.

From a prudent distance, however, the cart's erratic movements can be tracked with minimal difficulty and danger. Use the plane's rudder to spray bullets and missiles into a cloud about the vehicle. Shopping carts are not very tough, and in just a few lucky hits they will fall over dead.

Lead the top-notch carts into traps:

The top-notch shopping carts will follow your plane with uncanny

General strategy and flying hints (continued)

ability. It is almost impossible to get a clean shot at them because they are always directly underneath your plane.
  If they are the variety which do NOT have the flashing antenna, then your best bet is to fly directly over a fence, tick, or other structure in order
to lead them to their own destruction.

Climb and shoot:

  The high-quality flashing antenna carts can be defeated by climbing high,
dropping the flaps, and then come straight down at them with guns blazing.
Make sure to break off before you pile into the ground.

Vertical bombing:

  Ordinarily, bombs are not effective against the agile carts. However, in
the case of the high-quality models, you can use their tenacity against them.
Fly exactly straight up. Your pitch gauge should show your plane as being
straight up. Then drop three bombs, and remain vertical until the bombs explode. If you are lucky, the shopping cart will have stayed directly below you doing a little dance while the bombs rained down.
  Note that if you stall and fall out of your straight up climb before the
bombs reach the ground, then the shopping carts will have followed you and moved out of the target area by the time the bombs fall.

Attack shopping cart sites first:

  As with KLA's, it is a losing battle to try take out all the carts first, and then the sites. Instead, ignore the carts until all the sites
are history. Then have fun tearing up the panicked vehicles.

Tips for survival outside the plane

Get away from the plane after leaving it:

  When you are outside your plane, you are very vulnerable to enemy weapons. Contact with a KLA, or AAA shell results in instant death.
Since both weapons will attack your plane, not yourself, it is a good idea to get far away from your plane very quickly after ejecting.

Keep moving at all times if there are shopping carts around:

  The shopping carts are like the KLA's in that they always head directly
toward (despite their erratic motions) you. IE unlike the AAA, they do not lead. Therefore if you run in a straight direction, they will always
end up chasing behind you. If you stay put, they will wander in from all
directions until you have no direction which is safe.

General strategy and flying hints (continued)

Look down at your feet when jumping platforms:

When hopping from platform to platform, tilt your head down 30 to 40 degrees from the horizontal. Otherwise when you jump, you will not be able to see the platform you are attempting to jump onto. Jump high into
the air, and hold and release the W key to move forward just the right amount to land you on target.

Watch the far edge of the platform you are jumping to. As long as you don't pass over that edge, you will make the jump. Keep that far edge within your view, and you should have no troubles.

Use proper vertical angle with the handgun:

Don't always assume that you should aim for the center of an object when shooting with the handgun. Some targets take more hits with the game aimed high. If you see grey spheres appear where the bullets hit, then the bullets are striking the ground before damaging the target.

Experiment with position if assassin bomb fails:

The object-object collision algorithms used in Corncob were designed for fast-moving objects colliding with stationary targets. Assassin bombs will not always detonate their targets correctly because of the design of the game engine. For instance, if a mind bender is sitting on a force field, then a tricky situation can result. Both the force field and the mind bender can detonate a bomb. Therefore, when the detonate command is given, either the force field or the mindbender can be the cause of the bomb's detonation. If by chance the force field detonates the bomb first, then the mindbender will remain intact through the bomb's detonation.

Which object causes detonation is determined by the exact positioning of the bomb relative to the target and the person. Before collision processing, all objects are sorted by distance from the observer, then objects which are close together in the sorted list are compared for collision.

So if your bomb fails, try the next bomb in a position closer or farther away from the target. Make sure the center of the force field platform is not approximately the same distance as from you to the target, etc.

Use parachute to get extra hopping distance:

Your parachute can be used even when jumping into the air. If you are particularly nervous about a jump, you may pull the chute string at the apogee of your jump, and then float down slowly. The disadvantage is that you will swing for a bit, which can be disorienting.